Practice for Quiz1: MA 5104, Department of Mathematics, IIT Madras

Course name : *Statistical Inference and Learning*

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1. A researcher wants to estimate the proportion p of households in a city that own a particular type of appliance. How large a random sample is required to estimate p to within 0.04 with confidence coefficient 0.95, if a previous study suggests that $p \approx 0.35$? What sample size would be required if no prior estimate of p were available? Use $z_{0.025} = 1.96$.

2. A population proportion is to be estimated with margin of error 0.03 and confidence coefficient 0.99. If there is no prior information about the population proportion, determine the required sample size. Use $z_{0.005} = 2.576$.

3. Let $X_1, \ldots, X_n$ be a random sample from a population with mean μ and variance σ^2. Consider the estimator

$$\hat{\mu} = \frac{X_1 + X_2 + X_3}{3}.$$

Find the bias and MSE of $\hat{\mu}$.

4. Suppose X and Y are independent random variables with

$$E(X) = 3\theta, \qquad \mathrm{Var}(X) = 4,$$

and

$$E(Y) = \theta, \qquad \mathrm{Var}(Y) = 9.$$

Consider the estimator

$$\hat{\theta} = \frac{X + 2Y}{5}.$$

Find its bias, variance, and MSE.

5. Let X and Y be independent random variables satisfying

$$E(X) = 2\theta, \quad \mathrm{Var}(X) = 5,$$

$$E(Y) = 4\theta, \quad \mathrm{Var}(Y) = 8.$$

Consider

$$\hat{\theta} = \frac{3X + Y}{10}.$$

Determine whether the estimator is unbiased. Find its MSE.

6. Two estimators of the same parameter have the following properties:

$$E(\hat{\theta}_1) = \theta, \qquad \mathrm{Var}(\hat{\theta}_1) = 4,$$

and

$$E(\hat{\theta}_2) = \theta + 1, \qquad \mathrm{Var}(\hat{\theta}_2) = 1.$$

Find the MSE of each estimator. Which estimator would you prefer according to MSE?

7. An estimator $\hat{\theta}$ has

$$E(\hat{\theta}) = \theta + 2, \qquad \mathrm{Var}(\hat{\theta}) = 3.$$

Find its bias and MSE. What happens to the MSE if the bias is reduced to zero while the variance remains unchanged?

8. A random sample of 25 observations from a normal population gives

$$\bar{x} = 48, \qquad s = 5.$$

Find a 95% confidence interval for the population mean μ. Assume the population variance is unknown. Use $t_{0.025,24} = 2.064$.

9. A random sample of 36 observations gives a sample mean of 72 and a sample standard deviation of 12. Construct a 90% confidence interval for the population mean. Assume the population is normal and use $t_{0.05,35} = 1.690$.

10. A machine produces metal rods whose lengths are normally distributed. A sample of 16 rods has mean length 10.4 cm and sample variance 0.25 cm^2. Construct a 95% confidence interval for the population variance σ^2. You may use

$$\chi^2_{0.975,15} = 27.488, \qquad \chi^2_{0.025,15} = 6.262.$$

11. Two independent normal populations have the following sample summaries:

$$n_1 = 12, \quad \bar{x}_1 = 25, \quad s_1^2 = 4,$$
$$n_2 = 10, \quad \bar{x}_2 = 22, \quad s_2^2 = 5.$$

Assuming equal population variances, construct a 95% confidence interval for $\mu_1 - \mu_2$. Use $t_{0.025,20} = 2.086$.

12. Two independent samples are taken from normal populations. The sample summaries are

$$n_1 = 15, \quad s_1^2 = 6, \qquad n_2 = 12, \quad s_2^2 = 3.$$

Construct a 90% confidence interval for σ_1^2/σ_2^2. Use

$$F_{0.05,14,11} = 2.72, \qquad F_{0.05,11,14} = 2.60.$$

13. A company wants to compare the average battery life of two types of batteries. Independent samples give

$$n_1 = 20, \quad \bar{x}_1 = 8.2, \quad s_1^2 = 1.6,$$
$$n_2 = 18, \quad \bar{x}_2 = 7.5, \quad s_2^2 = 1.4.$$

Assuming the two populations are normal with equal variances, construct a 90% confidence interval for $\mu_1 - \mu_2$. Based on the interval, state whether the data provide evidence that the first battery has a different average lifetime.

14. For a normal population, a random sample of 10 observations has sample variance $s^2 = 16$. Construct a 95% confidence interval for the population variance σ^2. Use

$$\chi^2_{0.975,9} = 19.023, \qquad \chi^2_{0.025,9} = 2.700.$$

15. Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with density

$$f(x;\theta) = \frac{1}{\theta}e^{-x/\theta}, \qquad x > 0, \quad \theta > 0,$$

and suppose $E(X) = \theta$.

Find the method-of-moments estimator of θ, and determine whether the estimator is unbiased.

16. Let $X_1, \ldots, X_n$ be a random sample from a geometric distribution with

$$P(X = x) = p(1-p)^{x-1}, \qquad x = 1, 2, \ldots, \quad 0 < p < 1,$$

and suppose

$$E(X) = \frac{1}{p}.$$

Find the method-of-moments estimator of p.

17. Let $X_1, \ldots, X_n$ be a random sample from a normal distribution with mean μ and known variance σ^2. Derive the maximum likelihood estimator of μ.

18. Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with density

$$f(x;\theta) = \theta e^{-\theta x}, \qquad x > 0, \quad \theta > 0.$$

Derive the maximum likelihood estimator of θ.

19. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $(0, \theta)$, where $\theta > 0$.

(a) Find the method-of-moments estimator of θ.

(b) Find the maximum likelihood estimator of θ.

20. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $(-\theta, \theta)$, where $\theta > 0$.

(a) Find the method-of-moments estimator of θ using the first moment.

(b) Find the maximum likelihood estimator of θ.

21. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $(\theta, \theta + 1)$, where $\theta \in \mathbb{R}$.

Find the maximum likelihood estimator of θ.

22. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $(0, 2\theta)$, where $\theta > 0$.

(a) Find the method-of-moments estimator of θ.

(b) Find the maximum likelihood estimator of θ.

23. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $(\theta - 1, \theta + 1)$, where $\theta \in \mathbb{R}$.

Find the maximum likelihood estimator of θ.

24. Let $X_1, \ldots, X_n$ be a random sample from a population with probability density function

$$f(x;\theta) = \frac{2x}{\theta^2}, \qquad 0 < x < \theta, \quad \theta > 0.$$

(a) Find the method-of-moments estimator of θ, given that

$$E(X) = \frac{2\theta}{3}.$$

(b) Find the maximum likelihood estimator of θ.

25. Let $X_1, \ldots, X_n$ be a random sample from a distribution with probability density function

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}, \qquad x > 0, \quad \theta > 0.$$

(a) Find the method-of-moments estimator of θ.

(b) Find the maximum likelihood estimator of θ.

26. Let $X_1, \ldots, X_n$ be a random sample from a Bernoulli distribution with parameter p, where $0 < p < 1$.

(a) Find the method-of-moments estimator of p.

(b) Find the maximum likelihood estimator of p.

27. Let $X_1, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, where σ^2 is known.

Find the maximum likelihood estimator of μ.

28. Let $X_1, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, where both μ and σ^2 are unknown.

Find the maximum likelihood estimators of μ and σ^2.

29. Let $X_1, \ldots, X_n$ be a random sample from a Poisson distribution with parameter $\lambda > 0$ and probability mass function

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots.$$

(a) Find the method-of-moments estimator of λ.

(b) Find the maximum likelihood estimator of λ.

30. Let $X_1, \ldots, X_n$ be a random sample with density

$$f(x;\theta) = \frac{3x^2}{\theta^3}, \qquad 0 < x < \theta, \quad \theta > 0.$$

You may use the fact that

$$E(X) = \frac{3\theta}{4}.$$

(a) Find the method-of-moments estimator of θ.

(b) Find the maximum likelihood estimator of θ.